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Even though we often explain first law of thermodynamics using gases, this law is universal and applies to liquids and solids also.

Note

Some book presents the first law of thermodynamics as $\Delta U = Q + W$. Here the work done by the system is taken as negative and work done on the system is positive. However both the conventions are correct and we can follow any one of the convention.

EXAMPLE 8.12

A person does 30 kJ work on 2 kg of water by stirring using a paddle wheel. While stirring, around 5 kcal of heat is released from water through its container to the surface and surroundings by thermal conduction and radiation. What is the change in internal energy of the system?

Solution

Work done on the system (by the person while stirring), W = -30 kJ = -30,000J

Heat flowing out of the system, Q = -5 kcal = $5 \times 4184\ J$ =-20920 *J*

Using First law of thermodynamics

$$\Delta U = Q - W$$

$$\Delta U = -20{,}920\ J - (-30{,}000)\ J$$

$$\Delta U = -20{,}920\ J + 30{,}000\ J = 9080\ J$$

Here, the heat lost is less than the work done on the system, so the change in internal energy is positive.

EXAMPLE 8.13

Jogging every day is good for health. Assume that when you jog a work of 500 kJ is done and 230 kJ of heat is given off. What is the change in internal energy of your body?

Solution



Work done by the system (body),

$$W = +500\ kJ$$

Heat released from the system (body),

$$Q = -230\ kJ$$

The change in internal energy of a body

$$= \Delta U = -230\ kJ - 500\ kJ = -730\ kJ$$

8.6.3 Quasi-static process

Consider a system of an ideal gas kept in a cylinder of volume V at pressure P and temperature T. When the piston attached to the cylinder moves outward the volume of the gas will change. As a result the temperature and pressure will also change because all three variables P,T and V are related by the equation of state PV = NkT. If a block of some mass is kept on the piston, it will suddenly push the piston downward. The pressure near the piston will be larger than other parts of the system. It implies that the gas is in non-equilibrium state. We cannot determine pressure, temperature or internal energy of the system until it reaches another equilibrium state. But if the piston is pushed very slowly such that at every stage it is still in equilibrium with surroundings, we can use the equation of state to calculate the internal energy, pressure or temperature. This kind of process is called quasi-static process.